ECON573 Problem Set 9

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2. ILS Chapter 8: Exercise 5

Majority vote: The prediction is red. This is because the proportion of results with P(Class is Red|X) >= 0.5 is 6/10, so class red is the majority predicted class.

Average Probability: The prediction is green since the average probability = (0.1 + 0.15 + 0.2 + 0.2 + 0.55 + 0.6 + 0.6 + 0.65 + 0.7 + 0.75)/10 = 0.45 < 0.5

3. ILS Chapter 8: Applied Exercise 7

From the plot below, we observe that as the number of trees (B) increases, test MSE plummets initially; as B becomes larger and larger, test MSE approximate a horizontal line, which signifies that increasing B will not lead to overfitting issues. In practice, we could choose a sufficiently large B for the test MSE to have settled down.

Also, indeed as m decreases from p to sqrt(p), test MSE becomes lower, which means that m = sqrt(p) is indeed a good choice to decorrelate bagged trees and lower the error rate.

```
library(randomForest)
```

```
## randomForest 4.7-1.1
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
library(ISLR2)

set.seed(1)
train <- sample(1:nrow(Boston), nrow(Boston)/2)

Boston = as.data.frame(Boston)
x_train = Boston[train, -13]
y_train = Boston[train, 13]
x_test = Boston[-train, -13]
y_test = Boston[-train, 13]

p1 = ncol(Boston)-1
p2 = p1/3
p3 = p1/2
p4 = sqrt(p1)
```

```
mod_p1 = randomForest(x_train, y_train, xtest = x_test, ytest = y_test,
    mtry = p1, ntree = 800)
mod_p2 = randomForest(x_train, y_train, xtest = x_test, ytest = y_test,
    mtry = p2, ntree = 800)
mod_p3 = randomForest(x_train, y_train, xtest = x_test, ytest = y_test,
    mtry = p3, ntree = 800)
mod_p4 = randomForest(x_train, y_train, xtest = x_test, ytest = y_test,
    mtry = p4, ntree = 800)

plot(1:800, mod_p1$test$mse, col = "red", ylab = "Test MSE", xlab = "ntree", type = "l", ylim = c(15, 4
lines(1:800, mod_p2$test$mse, col = "blue", type = "l")
lines(1:800, mod_p3$test$mse, col = "green", type = "l")
lines(1:800, mod_p4$test$mse, col = "orange", type = "l")
legend(550, 40, legend = c("m=p", "m=p/3", "m=p/2", "m=sqrt(p)"), col = c("red", "blue", "green", "oran
```

